

Practical Tools for Study Design and Inference

Chapter 3. Single-Stage Designs

Sunghee Lee

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1. Background Ideas

Sample Size and Allocation for Single-Stage Designs

- Sample size calculation for specified precision (e.g., SE , $CV^2(=RelVariance)$, MOE) for
 - Means, totals and proportions — One group
 - Differences — Two + groups
- Allocation in stratified sampling
- Inflating sample sizes for clustering
- Sample sizes with nonresponse and attrition
- Sample size calculation for specified power

Probability Sampling

- 4 requirements of a probability sample from a finite population:
 - ① A set of samples can be defined that are possible to obtain with the sampling procedure.
 - ② Each possible sample s has a known probability of selection, $p(s)$.
 - ③ Every element in the target population has a non-zero probability of selection.
 - ④ One set of sample elements is selected with the probability associated with the set. Weights for sample units can be computed that are intended to project the sample to the target population.
- Statistical properties in design-based inference are computed wrt to the distribution of target estimates generated by repeated sampling.
 - Imagine that samples are selected over and over using a specified design. An estimate is computed from each sample.
 - Properties like the estimator's expected value and variance are computed across all the possible repeated samples.
 - This is different from the model-based inference you learn in most stat courses!

Terminology - 1

Population values

- U : Universe or population of size N
- Population mean: $\bar{y}_U = \frac{\sum_{i=1}^N y_i}{N}$
- Population (or unit or element) variance: $S^2 = \frac{\sum_{i=1}^N (y_i - \bar{y}_U)^2}{(N - 1)}$
- Population relvariance: $CV_U^2 = \frac{S^2}{\bar{y}_U^2}$
- Population coefficient of variation (CV): $CV_U = \frac{S}{\bar{y}_U}$

Terminology - 2

Sample estimates

- If $\hat{\theta}$ is an estimate of a population value, θ , then its standard error is $SE(\hat{\theta}) = \sqrt{V(\hat{\theta})}$, where V is the variance computed with respect to a particular sample design.
- $v(\hat{\theta})$ is an estimator of $V(\hat{\theta})$; *estimated variance* (Very different than S^2)
- $cv(\hat{\theta}) = \frac{\sqrt{v(\hat{\theta})}}{\hat{\theta}}$ is an estimator of $CV(\hat{\theta})$; *estimated coefficient of variation*; cv is unitless
- People often say "variance" or "standard error", when they mean "estimated variance" or "estimated standard error"

Criteria for Determining Sample Sizes

- Standard error (SE) of an estimate — Setting a target SE requires judgment to be made about an acceptable level of SE . Difficult because an SE has the same units as the analysis variable (e.g., persons, dollars, milligrams of mercury, etc.).
- Coefficients of variation — CV 's more useful than SE 's because they have no units of measure. Target values can be set without regard to the scale of an analysis variable.
- Margin of error (MOE) — Related to the width of a confidence interval. Useful because survey sponsors or analysts are often comfortable making statements like "I want to be able to say that the population value is within 3% of the sample estimate."
- $SE(\hat{\theta})$ vs. $\frac{SE(\hat{\theta})}{\hat{\theta}}$ vs. $z_{1-\alpha/2}SE(\hat{\theta})$

2. Sample Sizes: Simple Random Sample (SRS)

Definition of a Simple Random Sample

- Select a sample of n from a population of N
- There are $\binom{N}{n}$ combinations (sets) that can be selected without replacement.

$$\binom{N}{n} = \frac{N(N-1)\cdots(N-n+1)}{n(n-1)\cdots 1} = \frac{N!}{n!(N-n)!}$$

- In a simple random sample without replacement (*srswor*), each of the $\binom{N}{n}$ samples has the same chance of selection.
- The selection probability of any element is fixed at $\frac{n}{N}$.
- The weight of any element is fixed at $\frac{N}{n}$.
- A sample in which every element has the same weight is called **self-weighting**.

2.1. Sample Size for CV Target in $srswor$

Sample for CV target

- Estimated mean and total in srs

$$\bar{y}_s = \frac{\sum_{i \in s} y_i}{n}; \quad \hat{t} = N\bar{y}_s \quad (3.2)$$

- Variance of a total in an srs

$$V(\hat{t}) = N^2 \left(1 - \frac{n}{N}\right) \frac{S_U^2}{n} = N \left(\frac{N}{n} - 1\right) S_U^2$$

- Square of the coefficient of variation for $\bar{y}_s$ and $\hat{t}$

$$CV^2(\bar{y}_s) = \frac{V(\bar{y}_s)}{\bar{y}_s^2} = \left(\frac{1}{n} - \frac{1}{N}\right) \frac{S_U^2}{\bar{y}_s^2} \quad (3.3)$$

- Set CV^2 (i.e. relvariance) to some desirable value, say CV_0^2 (like 0.05), and solve for required sample n :

$$n = \frac{\frac{S_U^2}{\bar{y}_U^2}}{CV_0^2 + \frac{S_U^2}{N\bar{y}_U^2}} \quad (3.4)$$

Setting CV_0

- Somewhat arbitrary. Different surveys have different standards.
- If $CV = 50\%$, a normal approximation CI would cover 0
⇒ Bad estimate
- US Consumer Price Index (CPI) has SE target of 0.04 percentage points on 1-month change.
- Council of EU specifies that labor force estimates have $CV \leq 8\%$.
- US IRS requires CV of $0.06 (= 0.10/1.645)$ for some tax return estimates. This comes from margin of error in a 1-sided 95% CI.

See **VDK Example 3.1 in Class1Lab.pdf**

Estimating Proportions

- Estimate proportion of units that have some characteristic. Code $y_i = 1$ if unit i has characteristic and $y_i = 0$ if not.

$$p_s = \sum_{i \in s} y_i / n$$

- The unit relvariance is then defined as

$$\frac{S^2}{\bar{y}^2} = \frac{N}{N-1} \frac{p_U q_U}{p_U^2} = \frac{N}{N-1} \frac{q_U}{p_U} \doteq \frac{q_U}{p_U}$$

- The relvariance of p_s is

$$CV^2(p_s) = \left(\frac{1}{n} - \frac{1}{N} \right) \frac{N}{N-1} \frac{q_U}{p_U}$$

- Sample size that will achieve CV_0 (i.e., a target CV) comes from specializing earlier expression (3.4):

$$n = \frac{\frac{N}{N-1} \frac{q_U}{p_U}}{CV_0^2 + \frac{1}{N-1} \frac{q_U}{p_U}} \quad (3.8)$$

See **Example 3.4 in Class1Lab.pdf** after Fig 3.1

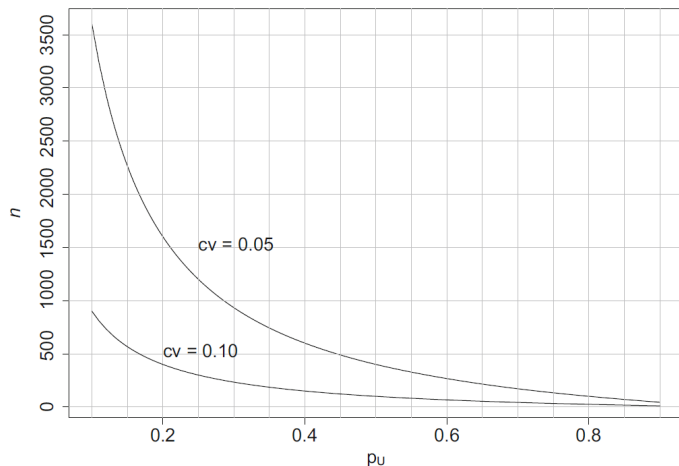
Fig 3.1: Sample sizes for $CV_0 = 0.05$ and $CV_0 = 0.10$ 

Fig. 3.1: Approximate sample sizes from Eq. (3.8) required to achieve CV s of 0.05 and 0.10 for population proportions ranging from 0.10 to 0.90. The population size is assumed to be large so that the finite population correction is one

2.2. Margin of Error for Setting Sample Sizes

Setting a Margin of Error - 1

- Many investigators prefer to think of setting a tolerance or *margin of error* rather than a *CV*.
- Margin of error (MOE)**. Suppose goal is to be within e of the population mean with probability $1 - \alpha$. Assuming that $\bar{y}_s$ can be treated as being normally distributed, set the half-width of a $100(1 - \alpha)\%$ two-sided CI to $e = z_{1-\alpha/2} \sqrt{V(\bar{y}_s)}$. This translates to

$$Pr(|\bar{y}_s - \bar{y}_U| \leq e) = 1 - \alpha \quad (3.10)$$

- If we set (i.e., target) the margin of error to e_0 above (3.10), then the required sample size is:

$$n = \frac{z_{1-\alpha/2}^2 S^2}{e_0^2 + z_{1-\alpha/2}^2 S^2 / N} \quad (3.12)$$

Setting a Margin of Error - 2

- **Margin of error relative to population mean.** Assuming that $\bar{y}_s$ can be treated as being normally distributed, set the half-width of a $100(1 - \alpha)\%$ two-sided CI to $e = z_{1-\alpha/2}CV(\bar{y}_s)$. This translates to

$$Pr\left(\left|\frac{\bar{y}_s - \bar{y}_U}{\bar{y}_U}\right| \leq e\right) = 1 - \alpha \quad (3.11)$$

- If we set the margin of error in (3.11) to e_0 , then the required sample size is:

$$n = \frac{z_{1-\alpha/2}^2 S^2 / \bar{y}_U^2}{e_0^2 + z_{1-\alpha/2}^2 S^2 / (N \bar{y}_U^2)} \quad (3.13)$$

See VDK Example 3.6 in Class1Lab.pdf

Other Methods for Proportions

- If p_s is extreme, a normal approximation CI can have an endpoint outside $[0, 1]$
- Two methods for getting CI's within bounds are Wilson and log-odds which result in asymmetrical CI. (Refer to the textbook for the content and R code.)
- These can be used to compute sample sizes, too, but results are usually similar to ones based on normal approximation intervals.

3. Sample Sizes: Stratified SRS (STSRS)

Mean & Variance for Stratified Sampling

- Stratification: divide population into mutually exclusive and exhaustive strata, select a sample from each
- The estimated mean in a *stsrswor* for the population with H strata is

$$\bar{y}_{st} = \sum_{h=1}^H W_h \bar{y}_{s_h} \quad (3.18)$$

- The variance estimator of the stratified sampling is

$$V(\bar{y}_{st}) = \sum_{h=1}^H W_h^2 \frac{1 - f_h}{n_h} S_h^2 \quad (3.20),$$

$$\text{where } W_h = \frac{N_h}{N}$$

Types of Stratum Allocations

- Fixed-cost methods
 - Proportional: $n_h = nW_h$, where $W_h = N_h/N$
 - Equal: $n_h = \frac{n}{H}$
 - Neyman: minimize $V(\bar{y}_{st})$
 - Cost-constrained optimal: minimize $V(\bar{y}_{st})$ subject to fixed total cost
- Fixed-variance method
 - Variance-constrained optimal: minimize total cost subject to fixed variance target, $V(\bar{y}_{st})$ or $CV(\bar{y}_{st})$

Neyman Allocation

- Suppose the total sample size is fixed. Allocation of *stsrswor* that minimizes $V(\bar{y}_{st})$ is

$$n_h = n \frac{W_h S_h}{\sum_{h=1}^H W_h S_h}$$

- More sample goes to strata that are a larger proportion of population or have higher unit standard deviations; i.e., larger $W_h \rightarrow$ larger n_h ; larger $S_h \rightarrow$ larger n_h
- Neyman allocation ignores costs

Allocation Considering Costs

- If costs are used, optimal allocation can be computed to
 - ① minimize variance subject to fixed total cost or
 - ② minimize cost subject to fixed variance (or *CV*) target
- Cost function: $C = c_0 + \sum_{h=1}^H n_h c_h$, where
 - C : total cost
 - c_0 : sum of cost that *do not vary* with the number of sample cases
 - c_h : the cost per sample case in stratum h
- c_0 is usually called "fixed cost" and can include components such as salaries for a project manager, programmers, and supervisors. The term c_h is the cost of data collection, e.g., interviewing and mailing, and other unit costs that increase as the sample size increases.

Allocation for Fixed Total Cost

- Minimizing $V(\bar{y}_{st})$ subject to a specified total budget leads to:

$$n_h = (C - c_0) \frac{W_h S_h / \sqrt{c_h}}{\sum_{h=1}^H (W_h S_h \sqrt{c_h})} \quad (3.21)$$

- The total sample size is the sum of the n_h across the sampling strata:

$$n = \sum_{h=1}^H n_h = (C - c_0) \frac{\sum_{h=1}^H (W_h S_h / \sqrt{c_h})}{\sum_{h=1}^H (W_h S_h \sqrt{c_h})} \quad (3.22)$$

- The proportion of the sample allocated to stratum h is:

$$\frac{n_h}{n} = \frac{W_h S_h / \sqrt{c_h}}{\sum_{h=1}^H (W_h S_h / \sqrt{c_h})} \quad (3.23)$$

Allocation for Fixed Variance or CV Target

- If $V(\bar{y}_{st})$ is fixed at V_0 and total cost is minimized, allocation to stratum h is

$$n_h = (W_h S_h / \sqrt{c_h}) \frac{\sum_{h=1}^H W_h S_h \sqrt{c_h}}{V_0 + N^{-1} \sum_{h=1}^H W_h S_h^2} \quad (3.24)$$

- If CV of $\bar{y}_{st}$ is fixed at CV_0 , this implies that V_0 should be set to $V_0 = CV_0^2 \times \bar{y}_U^2$. In this case, the proportion of the sample allocated to stratum h is also given by (3.23) but the total sample size is

$$n = \sum_{h=1}^H (W_h S_h / \sqrt{c_h}) \frac{\sum_{h=1}^H W_h S_h \sqrt{c_h}}{V_0 + N^{-1} \sum_{h=1}^H W_h S_h^2} \quad (3.25)$$

See VDK Example 3.9 in Class1Lab.pdf

4. Sample Sizes: Sampling with Varying Probabilities

Sampling with Varying Probabilities

- Sampling units in proportion to some measure of size (MOS) — extremely efficient in single-stage sampling for estimating totals if the MOS is closely related to y
- p_i : relative size of unit i
- Example: MOS in hospital population is number of beds, x_i

Relative size of hospital i is $p_i = x_i / \sum_U x_i$

- If a fixed size sample of n units is selected without replacement, selection probability is $\pi_i = np_i$. Method of sampling when MOS is x is called $pp(x)$ sampling
- The π -estimator of a mean, assuming that N is known, is defined in general as

$$\hat{y}_\pi = N^{-1} \sum_{i \in s} \frac{y_i}{\pi_i}$$

Using a Model - 1

- Special case of $\pi_i = np_i$: π -estimator is

$$\hat{y}_\pi = N^{-1} \sum_{i \in s} \frac{y_i}{np_i}$$

- If each y_i were exactly proportional to x , say $y_i = \beta x_i$, then π -estimator reduces to $\hat{y}_\pi = \beta \bar{x}_U$ in every sample.

But, with $y_i = \beta x_i$, the population mean of y is $\beta \bar{x}_U$; so, $\hat{y}_\pi$ would be perfect in every sample.

- Less restrictively, suppose y_i follows this model:

$$E_M(y_i) = \beta x_i$$

$$V_M(y_i) = v_i \quad (3.29)$$

Using a Model - 2

- $\hat{y}_\pi$ is model unbiased in the sense that $E_M(\hat{y}_\pi - \bar{y}_U) = 0$
- Good practice when constructing estimators is to do some modeling to determine whether there are any covariates that can be used as measures of size and to create estimators with lower variance than the simple π -estimator
- Godambe & Joshi (1965) showed that under model (3.29) the most efficient MOS for *pps* sampling is proportional to $\sqrt{v_i}$. Minimizes anticipated sampling variance.
- Assumes population total is estimated by π -estimator and model holds with independent errors.
- Isaki & Fuller (1982) extended this to a linear model where $E_M(y_i) = \mathbf{x}_i^T \boldsymbol{\beta}$ and $V_M(y_i) = v_i$

Finding MOS

- Model that may fit some establishment or institutional populations has variance with the form, $V_M(y_i) = \sigma^2 x_i^\gamma$, where x_i is an MOS and γ is a power.
- Typical values of γ are in the interval $[0, 2]$.
- Given specification of $E_M(y_i) = \mathbf{x}_i^T \boldsymbol{\beta}$, γ can be estimated iteratively.
 - Model, $y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i$, is fitted by OLS
 - Residuals, e_i , calculated.
 - Squared residual estimates $E_M(e_i^2) \doteq V_M(y_i)$
 - Slope in a regression of $\log(e_i^2)$ on $\log(x_i)$ is an approximate estimate of γ : $\log(e_i^2) = \gamma_0 + \gamma_1 \log(x_i) + \varphi$
 - What is the point here? Finding an MOS that minimizes sampling variance

Estimating γ

R function `gammaFit` estimates γ . Parameters in the function are:

<code>X</code>	matrix of predictors
<code>x</code>	vector of x 's in $V(Y)$
<code>Y</code>	vector of response variables
<code>maxiter</code>	maximum number of iterations allowed
<code>show.iter</code>	show values of gamma at each iteration, TRUE or FALSE
<code>tol</code>	relative change in gamma used to judge convergence

See **VDK Example 3.11** in **Class2Lab.pdf**

4.1. The With-Replacement Workaround

Variance in Without-Replacement *pps* Sampling

- The variance of $\hat{y}_\pi$ is complicated, because it involves joint selection probabilities of pairs of units

$$V(\hat{y}_\pi) = N^{-2} \sum_{i \in U} \sum_{j \in U} (\pi_{ij} - \pi_i \pi_j) \frac{y_i}{\pi_i} \frac{y_j}{\pi_j}$$

- π_{ij} is probability that units i and j are simultaneously in the sample.
- Not easy to use for sample size calculation

Calculations for With-Replacement Sampling - 1

- One practical approach is to use variance formula appropriate for *pps with replacement* (*ppswr*) sampling.
- Simple estimator of the mean studied with *ppswr* sampling is called "p-expanded with replacement" (Sändal, et al. 1992):

$$\hat{y}_{pwr} = \frac{1}{Nn} \sum_{i \in s} \frac{y_i}{p_i} \quad (3.31)$$

- A unit is included in the sum as many times as it is sampled.
- Although $\hat{y}_{pwr}$ looks just like $\hat{y}_{\pi}$, the selection probability of unit i is not np_i in with replacement sampling; it is actually:
 $1 - (1 - p_i)^n$.

Calculations for With-Replacement Sampling - 2

- The variance of $\hat{y}_{pwr}$ in *ppswr* sampling is

$$V(\hat{y}_{pwr}) = \frac{1}{N^2 n} \sum_{i \in U} p_i \left(\frac{y_i}{p_i} - t_U \right)^2 \equiv \frac{V_1}{N^2 n} \quad (3.32)$$

- We have n in (3.32). So we can solve for this.
- Advantage when computing a sample size is that n is clearly separated from the other terms, unlike in $V(\hat{y}_\pi)$
- If desired coefficient of variation is CV_0 , solve to give the sample size as

$$n = \frac{V_1}{N^2} \frac{1}{\bar{y}_U^2 CV_0^2} \quad (3.33) \quad CV^2 = \frac{V(\hat{y}_{pwr})}{\hat{y}_{pwr}^2} = \frac{1}{n}$$

20pt $pwr \frac{V_1}{N^2 n}$ – Solve for n to get to (3.33)

Calculations for With-Replacement Sampling - 3

- Note that (3.33) is like (3.4) with N large, if we interpret the population CV^2 to be $\frac{V_1}{N^2 \bar{y}_U^2} \rightarrow$ we can use **nCont** to compute a sample size.

$$(3.4) \quad n = \frac{\frac{S_U^2}{\bar{y}_U^2}}{CV_0^2 + \frac{S_U^2}{N \bar{y}_U^2}} = \frac{S^2}{CV_0^2 \bar{y}_U^2} = \frac{1}{CV_0^2} CV_{pop}^2$$

$$(3.33) \quad n = \frac{V_1}{N^2 \bar{y}_U^2} \frac{1}{CV_0^2} = \frac{1}{CV_0^2} CV_{pop}^2$$

- V_1 can be estimated from a sample that was selected with the same relative measures of size, p_i , as to be used in the planned sample.
- Or, it can also be estimated from a *pps* sample that was selected with some other MOS. (See VDK Ch. 3.4)

4.2. Accounting for Large Units

Example 3.12 Accounting for Large Units

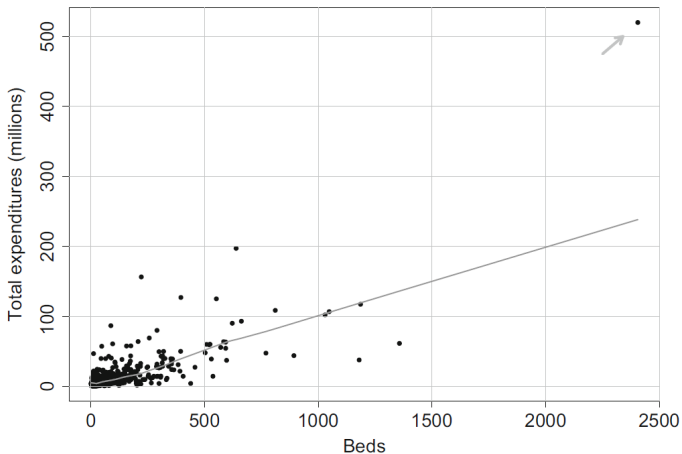


Fig. 3.3: Plot of total expenditures versus number of beds for the SMHO population. The *gray line* is a nonparametric smoother (lowess)

Example 3.12 Accounting for Large Units

- 671 hospitals in `smho98` with non-zero beds; point marked by arrow has 2,405 beds.
- Depending on sample size, large hospital will be selected with probability 1 in a *pps* sample with MOS x_i as $x_i \geq \frac{N\bar{x}_U}{n}$
- Sometimes relax to include all units with selection probabilities greater than some cutoff like 0.8.

In that case, the take-alls would be units with $x_i \geq 0.8 \frac{N\bar{x}_U}{n}$

- Take-all cutoffs depend on how big the sample is; the larger the sample size, the more units may be designated as take-alls.

Example 3.12 Accounting for Large Units

- Set aside big units and select *pps* sample from the remainder
- If we had n_t take-alls, the π -estimator of the mean will be

$$\hat{y}_\pi = N^{-1}[(N - n_t)\hat{y}_{\pi,nt} + T_{yt}],$$

where $(N - n_t)$ is the population counts of non-(take-all) units; $\hat{y}_{\pi,nt}$ is the π estimator of mean for non-(take-all) units; and T_{yt} is the total of the Y 's for the take-alls.

- For our example, the π -estimator is:

$$\hat{y}_\pi = N^{-1}[(N - 1)\hat{y}_{\pi,nt} + y_{2405}],$$

where $\hat{y}_{\pi,nt}$ is the π -estimator of mean for the $N - 1$ non-(take-all) units; and y_{2405} is the total for the unit with 2,405 beds.

- Since take-alls do not contribute to any sampling variability,

$$V(\hat{y}_\pi) = \left(\frac{N - n_t}{N}\right)^2 V(\hat{y}_{\pi,nt})$$

Sample Size with Certainties

- But the CV of $\hat{y}_\pi$ is still computed by dividing by $\bar{y}_U$:

$$CV(\hat{y}_\pi) = \frac{N - n_t}{N} \frac{\sqrt{V(\hat{y}_{\pi,nt})}}{\bar{y}_U}$$

- To calculate sample size, approximate $V(\hat{y}_{\pi,nt})$ by the *pwr* variance

$$V(\hat{y}_{\pi,nt}) \doteq \frac{V_1}{(N - n_t)^2 n}$$

V_1 refers only to subuniverse of $N - n_t$ non-(take-alls). The result is $V_1 = 9.53703e + 19$. Sample size formula is

$$n = \frac{V_1}{N^2} \frac{1}{\bar{y}_U^2 CV_0^2} = \frac{9.53703e + 19}{671^2 \times 13,667,706^2 CV_0^2}$$

with $\bar{y}_U = 13,667,706$. For $CV_0 = 0.15$, we get $n = 51$ for the non-(take-alls).

4.3. *pps* vs. Stratification

Relationship of *pps* Sampling to Stratification

- Stratification by size can be approx. same as *pp*(x) sampling
 - ➊ Sort frame from low to high based on the MOS.
 - ➋ Determine total sample size n
 - ➌ Divide sorted frame into $H = n/2$ strata such that the total of the MOS is about the same in each stratum.
 - ➍ Select *srswor* of size 2 in each stratum.
- If z_{hi} is MOS for unit i in stratum h and the MOS's do not vary much within a stratum, the selection probability in stratum h will be

$$\pi_{hi} = \frac{2}{N_h} \doteq \frac{2z_{hi}}{N_h \bar{z}_h}$$

where $\bar{z}_h$ is the average MOS in stratum h , and assuming $z_{hi} \doteq \bar{z}_h$.

Substitution for Nonresponse

- Substitution sometimes used for nonrespondents in surveys of institutions (schools, hospitals, prisons)
- Initial sample sizes may not be that large
- For each initially selected unit, select a substitute
- If unit refuses, then pre-selected substitute used
- If *pps* sampling used, every unit may have different selection probability
 - What weight does a substitute get?
- With *stsrswor* no ambiguity

See **VDK Example 3.13** in **Class2Lab.pdf**

4.4. Sample Sizes for Regression Estimators

Regression Estimates of Totals

- Models can be used to construct estimates of means and totals that are more efficient than π -estimators
- Suppose that the following linear regression model holds:

$$E_M(y_i) = \sum_{j=1}^p \beta_j x_{ji}; \quad V_M(y_i) = \sigma^2 v_i$$

- General regression estimator (GREG): $\hat{y}_r = \hat{y}_\pi + \sum_{j=1}^p b_j (\bar{x}_{Uj} - \hat{x}_{\pi j})$
where b_j is the estimate of β_j using survey-weighted least squares, $\bar{x}_{Uj}$ is the population mean of x_j , and $\hat{x}_{\pi j}$ is the π -estimator of the mean of x_j
- Slope estimator $\hat{\mathbf{B}} = (\mathbf{X}^T \mathbf{D} \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{D} \mathbf{V}^{-1} \mathbf{y}$ (More in Ch. 14)
- $\mathbf{V}$ is often set to the identity
- The GREG is model-unbiased in the sense that $E_M(\hat{y}_r - \bar{y}_U) = 0$ and is approximately design-unbiased
- The better the model fits, the lower the variance of the GREG

Optimal selection probabilities, i.e., minimizing the anticipated variance, are

$$\pi_i = \frac{nv_i^{1/2}}{N\bar{v}_U^{(1/2)}}$$

with $\bar{v}_U^{(1/2)} = \sum_U \sqrt{v_i}/N$. With these optimal probabilities, the approximate anticipated variance itself is

$$AV(\hat{y}_r) \doteq \left[\frac{1}{n} \left(N\bar{v}_U^{(1/2)} \right)^2 - N\bar{v}_U \right] \frac{\sigma^2}{N^2}, \quad (3.36)$$

where $\bar{v}_U = \sum_U v_i/N$. Dividing (3.36) by $[E_M(\bar{y}_U)]^2$, we get a kind of relvariance.

Setting the resulting relvariance above equal to CV_0^2 of r and solving for n leads to

$$n = \frac{\left[\bar{v}_U^{(1/2)} \right]^2}{CV_0^2 \frac{[E_M(\bar{y}_U)]^2}{\sigma^2} + \frac{\bar{v}_U}{N}} \quad (3.37)$$

See VDK Example 3.14 in Class2Lab.pdf

5. Estimating Population Parameters

Estimating Population Parameters from a Sample - 1

- If previous sample was selected in the same way as the planned sample, estimation is straightforward. If a different type of sample is planned from the earlier one, things are more complicated.
- *srswor*: Suppose earlier sample, s_0 , was an *srswor* of size n_0 . The unbiased estimators of the mean and element variance are:

$$\bar{y}_U = \sum_{i \in U} y_i / N \longrightarrow \bar{y}_{s_0} = \sum_{i \in s_0} y_i / n_0$$

$$S^2 = \frac{\sum_{i \in U} (y_i - \bar{y}_U)^2}{N - 1} \longrightarrow \hat{S}^2 = \frac{\sum_{i \in s_0} (y_i - \bar{y}_{s_0})^2}{n_0 - 1}$$

- In the special case of a binary y , the corresponding estimators are:

$$\bar{y}_{s_0} = p_0$$

$$\hat{S}^2 = \frac{n_0 p_0 (1 - p_0)}{n_0 - 1}$$

Estimating Population Parameters from a Sample - 2

- If sample was an *stsrswor*, then compute estimates within strata
- In some cases, we will have no microdata but an estimate of variance, $v(\bar{y}_{s_0})$, (or its square root) may be published. Assuming s_0 was an *srswor* of size n_0 , unit variance can be estimated as

$$\hat{S}^2 = \frac{n_0 v(\bar{y}_{s_0})}{1 - f_0}$$

where $f_0 = n_0/N$.

- If previous sample was more complex than *srswor*, but we have a design effect for the estimated mean, $\hat{y}$, then

$$\hat{S}^2 = \frac{n_0 v(\hat{y})}{1 - f_0} \frac{1}{\text{deff}(\hat{y})}$$

Estimating Population Parameters from a *pps* Sample

- *pps* sample of size n was selected using MOS's $\{p_i\}_{i=1}^N$. Even though s_0 was probably selected without replacement, standard work-around is to treat design as if it was *ppswr*. The estimate of the parameter V_1 is

$$\begin{aligned}\hat{V}_1 &= \frac{1}{n-1} \sum_{s_0} \left(\frac{y_i}{p_i} - \frac{1}{n} \sum_{s_0} \frac{y_i}{p_i} \right)^2 \\ &= \frac{n^2}{n-1} \sum_{s_0} \left(w_i y_i - \frac{1}{n} \sum_{s_0} w_i y_i \right)^2\end{aligned}\quad (3.41)$$

where $w_i = (np_i)^{-1}$.

Estimating Params from a *pps* Sample with new MOS

- If the plan is to select the new sample with another set of probabilities $\{q_i\}_{i=1}^N$, then the new V_1 can still be estimated. The new V_1 is

$$V_1 = \sum_U q_i \left(\frac{y_i}{q_i} - t_U \right)^2 = \sum_U \frac{y_i^2}{q_i} - t_U^2 \quad (3.42)$$

The term $\sum_U y_i^2/q_i$ is a population total and can be estimated by $n^{-1} \sum_{s_0} y_i^2/(q_i p_i)$. An unbiased estimator of (3.42) is

$$\hat{V}_1 = \frac{1}{n} \sum_{s_0} \frac{y_i^2}{q_i p_i} - \left(\frac{1}{n} \sum_{s_0} \frac{y_i}{p_i} \right)^2 + v(\hat{t}_\pi) \quad (3.43)$$

where $v(\hat{t}_\pi)$ is the estimated variance of $\hat{t}_\pi = n^{-1} \sum_{s_0} y_i/p_i$

Estimating Unit Variance

- If s_0 is selected with varying probabilities (not necessarily *pps*) and the inverse selection probabilities are $\{w_i\}_{i \in s_0}$, the unit variance parameter can also be estimated approximately as:

$$\hat{S}^2 = \frac{n}{n-1} \frac{\sum_{s_0} w_i (y_i - \bar{y}_w)^2}{\sum_{s_0} w_i - 1}$$

where $\bar{y}_w = \sum_{s_0} w_i y_i / \sum_{s_0} w_i$.

- This expression also applies to estimating stratum population variance, S_h^2 , based on the sample s_{0h} .
- $\hat{S}^2$ does have a negative bias, although the problem will be an issue only in small samples.

6. Rare Characteristics & Domain Estimates

Rare Characteristics

- For an extremely rare characteristic, there may be non-negligible probability of obtaining no cases at all
- Can find sample size so that probability of 0 sample cases is specified value
- Upper bound on number of rare cases: use hypergeometric distribution
- Upper bound on proportion rare: rule of 3
- If 0 rare observed, $3/n$ is an approximate upper bound on p_U

Domain Estimates - 1

- Kish's taxonomy of domains:
 - ① Design Domains: subpopulations that are restricted to specific strata (e.g., Ontario in a survey in Canada where provinces are strata);
 - ② Cross-classes: groups that are broadly distributed across the strata and primary sampling units (PSUs) (e.g., African Americans over the age of 50 in the U.S.);
 - ③ Mixed Classes: groups that are disproportionately distributed across the complex sample design (e.g., Hispanics in a sample that includes Los Angeles or Houston).
- The cells in any cross-tab define domains – age $\times$ gender, region $\times$ urbanicity.
- Unintended domains – out-of-business listings on frame $\rightarrow$ eligibles are a domain.

Domain Estimates - 2

- A goal of some surveys is to sample a few domains at higher rates than they occur in the population – known as **oversampling**.

If we want equal size samples of Whites and African-Americans in a household survey in U.S., we will have to sample the latter at a much higher rate than the former.

- Why not make each domain a design stratum so that the sample size in each can be controlled? Cannot always be done.
 - ① Frame may not give domain membership for all units in advance of sampling (e.g., adults looking for work)
 - ② Using the domains for strata may be impractical. The domains may not be disjoint.

We may want to analyze persons in domains defined by gender and race/ethnicity.

Complete gender $\times$ race/ethnicity cross could be too cumbersome to use as individual strata.

Domain Estimates & Systematic Sampling

- When domain identifiers are available on the frame but explicit strata using all domains are not formed, practitioners often try to ensure representation of each by using systematic sampling.
- Frame might be sorted by gender and then by race/ethnicity within gender. A systematic, equal probability sample would be distributed by gender and race/ethnicity much like the population.
- This method would usually eliminate samples that are poorly distributed among the domains but would not oversample any domain.

Sample Sizes for Domain Estimates

Two options in designing a sample to adequately cover analysis domains

- ➊ Calculate the expected numbers of units that will occur in the sample in each domain for a particular total sample size. Total sample size is then made large enough so that, in expectation, key domains of interest will be adequately represented. For example, according to the 2006 NHIS about 14.8% of persons did not have any type of health insurance. If an equal probability sample of persons were selected and 1,000 persons desired in the uninsured domain, we would need a sample of about 6,760 ($= 1,000/0.148$) to get 1,000 in expectation.
- ➋ Two-phase sample. Screen large, initial sample. Identify domain members. Subsample at rates computed to produce desired sample sizes.

Variance Formulas for Estimated Domain Totals - 1

- Suppose that *srswor* is selected and that domain membership is unknown before sampling is done.
- Estimate of a domain total for a variable y is $\hat{t}_d = (N/n) \sum_s y_{d_i}$ where y_{d_i} is the value of the variable for a unit if it is in domain d and is 0 if the unit is not in the domain. This can also be written as $y_{d_i} = y_i \delta_i$ with $\delta_i = 1$ if unit i is in the domain and 0 if not.
- The variance of $\hat{t}_d$ is

$$V(\hat{t}_d) = \frac{N^2}{n} \left(1 - \frac{n}{N}\right) S^2$$

where the unit variance is calculated including the zeros for non-domain units.

Variance Formulas for Estimated Domain Totals - 2

- Unit variance can be rewritten as $S^2 \doteq P_d(S_d^2 + Q_d\bar{y}_{U_d}^2)$, where S_d^2 is the variance among units that are in the domain, $\bar{y}_{U_d}$ is the population mean for those units, $P_d = N_d/N$ is the proportion of units in the population that are in the domain, and $Q_d = 1 - P_d$
- Using this version of S^2 , the relvariance of $\hat{t}_d$ is

$$CV^2(\hat{t}_d) \doteq \frac{1}{n} \left(1 - \frac{n}{N}\right) \frac{CV_d^2 Q_d}{P_d} \quad (3.46)$$

where $CV_d^2 = S_d^2/\bar{y}_{U_d}^2$ is the unit relvariance among the domain units.

- Setting equal to a target value CV_0^2 and solving for n gives

$$n = \frac{CV_d^2 + Q_d}{P_d CV_0^2 + \frac{CV_d^2 + Q_d}{N}} \doteq \frac{CV_d^2 + Q_d}{P_d CV_0^2} \quad (3.47)$$

Variance Formulas for Estimated Domain Means - 1

- An approximate variance is needed. Suppose that the mean is estimated by $\hat{y}_d = \hat{t}_d / \hat{N}_d$, where $\hat{N}_d = N n_d / n$.
- Linearly approximating $\hat{y}_d$ leads to

$$\hat{y}_d - \bar{y}_{U_d} \doteq \frac{1}{N} N \bar{e}_s,$$

where $\bar{e}_s = \sum_s e_i / n$ with $e_i = \delta_i(y_i - \bar{y}_{U_d})$.

- Approximate variance is

$$V(\hat{y}_d) \doteq \frac{1}{N_d^2} \frac{N^2}{n} \left(1 - \frac{n}{N}\right) S_e^2,$$

where $S_e^2 = (N - 1)^{-1} \sum_U e_i^2$

Variance Formulas for Estimated Domain Means - 2

- Using $e_i = y_i - \bar{y}_{U_d}$ for units in the domain $e_i = 0$ for non-domain units, we get $S_e^2 = (N - 1)^{-1} \sum_{U_d} (y_i - \bar{y}_{U_d})^2 \doteq P_d S_d^2$.
- Relvariance of $\bar{y}_{U_d}$ is then

$$CV(\bar{y}_{U_d})^2 = \frac{1}{nP_d} \left(1 - \frac{n}{N}\right) CV_d^2.$$

- Setting this equal to CV_0^2 and solving for n gives

$$n = \frac{CV_d^2}{P_d CV_0^2 + \frac{CV_d^2}{N}} \doteq \frac{CV_d^2}{P_d CV_0^2} \quad (3.48)$$

The sample size for estimating a mean can be substantially smaller than the one for estimating the domain total in (3.47).

Sample Sizes to Obtain $CV = 0.05$ for Totals and Means

P_d	n for total	n for mean
0.05	15,600	8,000
0.25	2,800	1,600
0.50	1,200	800
0.75	667	533
1.00	400	400

The population size is assumed to be large;
 domain relvariance is $CV_d^2 = 1$.

7. Design Effects

Design Effects

- Previous sample size calculations were for *srs* or *stsr*s
- Design effect is ratio of variance of estimator with design actually used to variance of estimator of same parameter with *srs*

$$\text{deff}(\hat{\theta}) = \frac{V(\hat{\theta})}{V_{srs}(\hat{\theta})}$$

- n/deff is **effective sample size**, i.e., the *srs* size that would produce the same variance
- A rough estimate of the total sample needed with a complex design (strata, clusters, *pps* sampling) can be found by computing the *srs* sample size, then multiplying by *deff*

$$\begin{aligned} n_{srs} \times \text{deff} &= (100 \text{ elements}) \times (\text{deff } 1.5) \\ &= 150 \text{ elements selected via complex design} \end{aligned}$$

More on deff's

- Computing sample size as n/deff is crude in some designs. In a 2-stage cluster sample, this tells you nothing about the split between clusters and elements per cluster.
- deff will not apply unless the new sample has the same numbers of clusters and elements per cluster as the one used to compute the deff .
- When using a deff from a computer package, make sure you understand how it was computed. Check to see whether *srs* variance was computed with or without an *fpc*.
- deff 's from a previous survey will not be useful if a new design is planned.
Strata and clusters may change. Subgroups may have target sample sizes so that an overall deff does not apply.

8. Software for Sample Selection

R packages and functions

Package	Function	Description
base	sample	Select <i>srswr</i> or <i>srswor</i> samples
pps	ppss	Systematic <i>ppswor</i> sampling
	ppssstrat	Stratified <i>ppswor</i> systematic sampling
	ppswr	<i>pps</i> sampling with replacement
	stratsrs	<i>stsrswor</i>
sampling	cluster	Single-stage cluster sampling
	srswor	Select <i>srswor</i> samples
	srswr	Select <i>srswr</i> samples
	strata	Select <i>stsrswor</i> , <i>stsrswr</i> , Poisson, and systematic samples
	UPrandomsystematic	Systematic <i>ppswor</i> sampling after randomizing the order of the list
	UPsampford	Sampford's method of <i>ppswor</i>
survey	stratsample	Select <i>stsrwor</i> samples

R sampling package

- Tillé & Matei (2012). *sampling*: Survey Sampling.
<http://CRAN.R-project.org/package=sampling>
- *pps* methods that allow joint selection probabilities to be computed
UPmaxentropy, UPmidzuno, UPsampford,
UPsystematic (Madow method), UPtille
- *pps* method that does not allow joint selection probabilities to be computed

See Example SRS Selection and VDK Example 3.18 in Class2Lab.pdf (This is not part of the homework assignment but should be helpful for the project.)